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Area and Volume of Similar Shapes

Length $\times k$, area $\times k^2$, volume $\times k^3$. Go back to the length scale factor first, then jump to what you need.

THE TOPIC ON ONE MAP

Area and Volume of Similar Shapes

01 Length, area and volume scale factors

k, k^2, k^3 ratio of areas

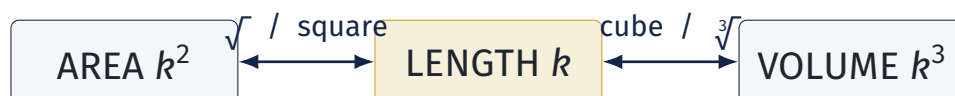
02 Find one from another

area $\rightarrow$ volume volume $\rightarrow$ area

03 Harder problems

sum or difference given frustum algebra

THE LADDER Always climb through the length scale factor:



01 Length, Area and Volume Scale Factors

Find k from one pair of matching lengths. Areas (including surface area) use k^2 ; volumes, masses and capacities use k^3 .

WHEN YOU SEE → YOU DO

two lengths given

$$\rightarrow k = \frac{\text{new}}{\text{old}}$$

“surface area”, “area of the base”

$$\rightarrow \times k^2$$

“volume”, “mass”, “how much paint to fill”

$$\rightarrow \times k^3$$

“ratio of the areas”

→ square the length ratio

WORKED MODEL • SMALL AND LARGE MONSTER

Heights 20 cm and 120 cm. The small one has surface area 300 cm². Large surface area?

$$1 \quad k = \frac{120}{20} = 6$$

length

$$\text{ANSWER} \quad 300 \times 6^2 = 10\,800 \text{ cm}^2$$

× **TRAP** Multiplying an area by k . Area needs k^2 , volume needs k^3 .

IDEA 01

Practise: Length, Area and Volume Scale Factors

My questions first, then selected practice.

Q1 **DR ESLAM** Area from length

CH 4 · TOPIC 9 2 marks

Two similar shapes have lengths in the ratio 2 : 5. The small one has area 12 cm^2 . Find the area of the large one.

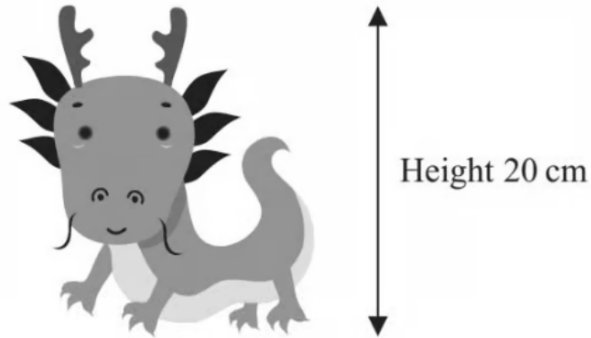
SELECTED PRACTICE

Q2 **PRACTICE** Practice · Medium

CH 4 · TOPIC 9 3 marks

1 A company makes monsters.

The company makes small monsters with a height of 20 cm.



A small monster has a surface area of 300 cm^2 .

The company also makes large monsters with a height of 120 cm.

A small monster and a large monster are mathematically similar.

Work out the surface area of a large monster.

(3 marks)

Q3**PRACTICE**

Practice • Medium

CH 4 • TOPIC 9 2 marks

- 9 A and B are similar solid cylinders.

base area of A : base area of B = 9 : 25

Complete these ratios.

curved surface area of A : curved surface area of B = :

height of A : height of B = :

(2 marks)

Q4**PRACTICE**

Practice • Medium

CH 4 • TOPIC 9 3 marks

- 14 A transport lorry consists of a cab and a trailer.
The trailer has a volume of 90m^3 .
Alfie makes a model of this lorry using a scale of 1 : 72.

Work out the volume of the trailer in Alfie's model, giving your answer in cm^3 .

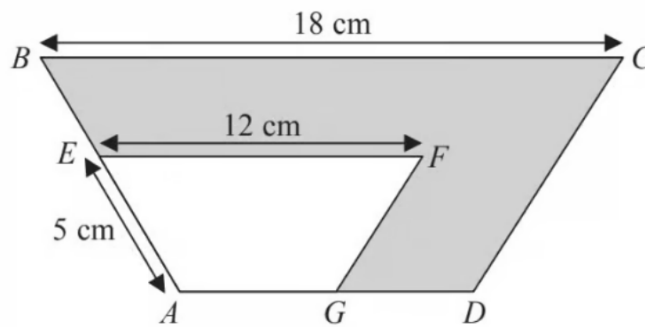
..... cm^3

(3 marks)

Q5**PRACTICE**

Practice · Hard

CH 4 · TOPIC 9 5 marks

3 (a)Diagram **NOT**
accurately drawn

$AE = 5 \text{ cm}$
 $ABCD$ and $AEFG$ are mathematically similar trapeziums. $EF = 12 \text{ cm}$
 $BC = 18 \text{ cm}$

Work out the length of AB .**(2 marks)****(b)** Trapezium $AEFG$ has an area of 36 cm^2 .

Work out the area of the shaded region.

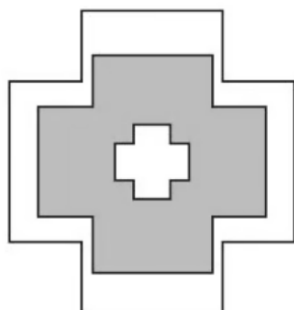
(3 marks)

Q6**PRACTICE**

Practice · Hard

CH 4 · TOPIC 9 4 marks

- 14** The diagram consists of three mathematically similar shapes. The heights of the shapes are in the ratio 1 : 4 : 5.



Not to scale

Find the ratio

total shaded area : total unshaded area.

Give your answer in its simplest form.

total shaded area : total unshaded area :

(4 marks)

Q7**PRACTICE**

Practice · Very Hard

CH 4 · TOPIC 9 2 marks

- 7 The area of pentagon A is 73.5 cm^2 .
The area of pentagon B is 6 cm^2 .

Find the ratio perimeter of pentagon A : perimeter of pentagon B in its simplest form.

..... :
(2 marks)

02 Find One from Another

Given areas, square-root to get k . Given volumes, cube-root to get k . Then raise k to the power you need.

WHEN YOU SEE → YOU DO

surface areas given, volume wanted

→ $k = \sqrt{\text{area ratio}}$, then k^3

volumes given, surface area wanted

→ $k = \sqrt[3]{\text{volume ratio}}$, then k^2

“surface area of A : surface area of B
= 25 : 36”

→ $k = \frac{6}{5}$

three solids A, B, C

→ convert every ratio to lengths, then link

Ratio given

Root back to k

Power you need

WORKED MODEL · MAY/NOV 2020 P1H

R: surface area 108, volume 135. S: surface area 300. Volume of S?

$$1 \quad k^2 = \frac{300}{108} = \frac{25}{9} \Rightarrow k = \frac{5}{3}$$

area → length

ANSWER $135 \times \left(\frac{5}{3}\right)^3 = 625 \text{ cm}^3$

IDEA 02

Practise: Find One from Another

My questions first, then selected practice, then real past papers.

Q8**DR ESLAM**

Volume from area

CH 4 · TOPIC 9

3 marks

Two similar solids have surface areas 36 cm^2 and 81 cm^2 . The small one has volume 40 cm^3 . Find the large volume.

SELECTED PRACTICE

Q9**PRACTICE** Practice · Medium

CH 4 · TOPIC 9 3 marks

7 The diagram shows two mathematically similar vases, **A** and **B**.

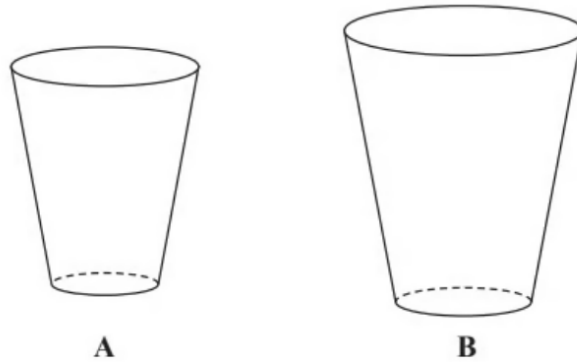


Diagram **NOT**
accurately drawn

A has a volume of 405cm^3

B has a volume of 960cm^3

B has a surface area of 928cm^2

Work out the surface area of **A**.

(3 marks)

Q10**PRACTICE**

Practice · Medium

CH 4 · TOPIC 9 3 marks

11 Solids **X** and **Y** are similar.

X has volume 64 cm^3

Y has volume 343 cm^3

The surface area of **X** is 176 cm^2

Work out the surface area of **Y**.

..... cm^2

(3 marks)

Q11**PRACTICE**

Practice · Hard

CH 4 · TOPIC 9 3 marks

5 Solid **A** and solid **B** are mathematically similar.

The ratio of the surface area of solid **A** to the surface area of solid **B** is 4: 9

The volume of solid **B** is 405 cm^3 .

Show that the volume of solid **A** is 120 cm^3 .

(3 marks)

Q12**PRACTICE**

Practice · Hard

CH 4 · TOPIC 9

3 marks

7 Cone **A** and cone **B** are mathematically similar.

The ratio of the volume of cone **A** to the volume of cone **B** is 27:8

The surface area of cone **A** is 297 cm^2

Show that the surface area of cone **B** is 132 cm^2

(3 marks)**PAST PAPERS****Q13****PAST PAPER**

Edexcel 4MA1 · Oct/Nov 2020 · Paper 1H · Q20

CH 4 · TOPIC 9

3 marks

20 **R** and **S** are two similar solid shapes.

Shape **R** has surface area 108 cm^2 and volume 135 cm^3

Shape **S** has surface area 300 cm^2

Work out the volume of shape **S**.

..... cm^3

(Total for Question 20 is 3 marks)

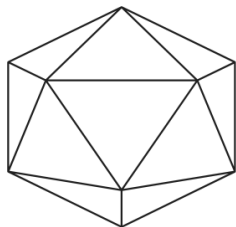
Q14

PAST PAPER

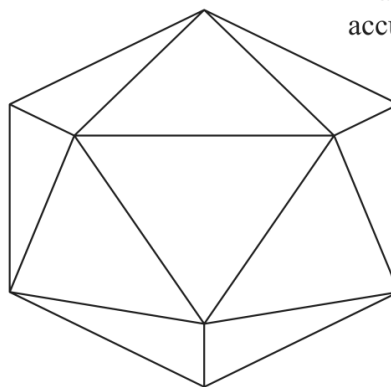
Edexcel 4MA1 · January 2021 · Paper 1H · Q20

CH 4 · TOPIC 9 3 marks

20 **A** and **B** are two similar solids.



A



B

Diagram **NOT**
accurately drawn

A has a volume of 1836 cm^3

B has a volume of 4352 cm^3

B has a total surface area of 1120 cm^2

Work out the total surface area of **A**.

..... cm^2

(Total for Question 20 is 3 marks)

Q15**PAST PAPER**

Edexcel 4MA1 · May 2021 · Paper 2H · Q20

CH 4 · TOPIC 9 3 marks

20 Mathematically similar wooden blocks are made in a workshop.

There are small blocks and there are large blocks.

The volume of each small block is 300 cm^3

Given that

the surface area of each small block : the surface area of each large block = $25 : 36$

work out the volume of each large block.

..... cm^3

(Total for Question 20 is 3 marks)

Q16**PAST PAPER**

Edexcel 4MA1 · January 2022 · Paper 1HR · Q18

CH 4 · TOPIC 9 4 marks

18 The three solids **A**, **B** and **C** are similar such that

the surface area of **A** : the surface area of **B** = 4 : 9

and

the volume of **B** : the volume of **C** = 125 : 343

Work out the ratio

the height of **A** : the height of **C**

Give your ratio in its simplest form.

.....
(Total for Question 18 is 4 marks)

Q17**PAST PAPER**

Edexcel 4MA1 · June 2022 · Paper 1HR · Q20

CH 4 · TOPIC 9 4 marks

20 The diagram shows two similar metal statues.

**A****B**

Diagram **NOT**
accurately drawn

The volume of statue **B** is 20% less than the volume of statue **A**

The surface area of statue **B** is $k\%$ less than the surface area of statue **A**

Work out the value of k

Give your answer correct to 3 significant figures.

$k =$

(Total for Question 20 is 4 marks)

Q18**PAST PAPER**

Edexcel 4MA1 · June 2024 · Paper 2HR · Q24

CH 4 · TOPIC 9 5 marks

24 The diagram shows a solid, **S**, made from a cone and a hemisphere.

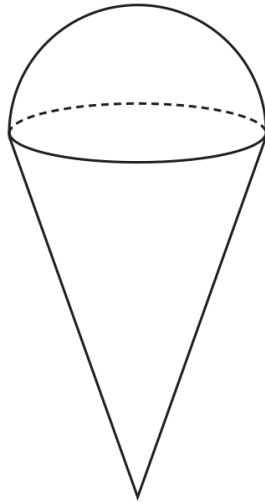


Diagram **NOT**
accurately drawn

The centre of the circular face of the cone coincides with the centre of the flat surface of the hemisphere.

The radius of the circular face of the cone, x cm, is equal to the radius of the hemisphere.

The total height of **S** is $4 \times$ the radius of the hemisphere.

A separate sphere has radius kx cm.

The volume of this sphere is $12.5 \times$ the volume of **S**

(a) Work out the value of k

$k = \dots\dots\dots$

A solid, **T**, is similar to solid **S**

The volume of **T** is $512 \times$ the volume of **S**

The total surface area of **T** is $d \times$ the total surface area of **S**

(b) Find the value of d

$d = \dots\dots\dots$

(Total for Question 24 is 5 marks)

Q19**PAST PAPER**

Edexcel 4MA1 · November 2025 · Paper 2H · Q23

CH 4 · TOPIC 9 3 marks**23** Shape **P** is similar to shape **Q**The table shows some information about shape **P** and shape **Q**

	Surface area (cm ²)	Volume (cm ³)
Shape P	200	672
Shape Q	450	

Work out the volume of shape **Q**..... cm³**(Total for Question 23 is 3 marks)**

03 Harder Problems: Differences, Frustums, Algebra

When a sum or difference is given, call the small one V and the large one k^3V , then form one equation. A frustum is the large cone minus the small cone.

WHEN YOU SEE → YOU DO

“the difference between the volumes is 1197”

$$\rightarrow k^3V - V = 1197$$

“surface area A + surface area B = 1800”

$$\rightarrow A + k^2A = 1800$$

a frustum of a cone or a pyramid

→ large – small, k from the heights

powers like $3^x, 8^x$

→ use k from areas, then index laws

WORKED MODEL · JANUARY 2022 P1H

Vases of heights 10 cm and 15 cm; the volumes differ by 1197 cm^3 . Find the volume of A.

1 $k = 1.5, k^3 = 3.375$

scale factors

2 $3.375V - V = 1197$

difference

ANSWER $V = 504 \text{ cm}^3$

IDEA 03

Practise: Harder Problems: Differences, Frustums, Algebra

My questions first, then selected practice, then real past papers.

Q20 **DR ESLAM** Difference

CH 4 • TOPIC 9 3 marks

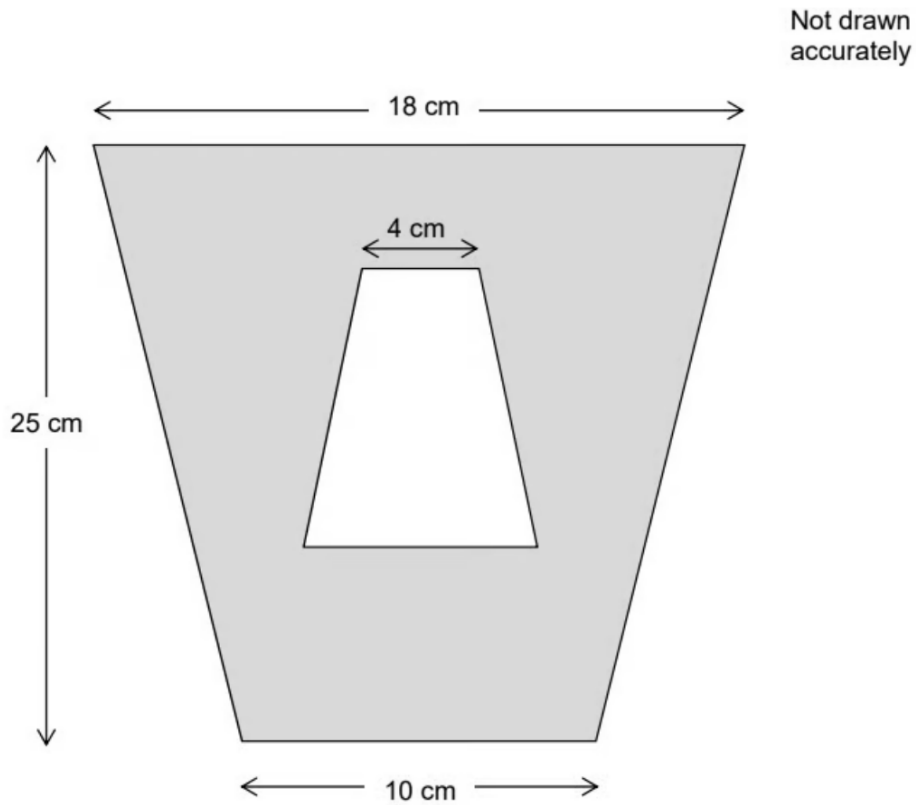
Two similar jugs have heights 8 cm and 12 cm. Their volumes differ by 570 cm^3 . Find the small volume.

SELECTED PRACTICE

Q21 PRACTICE Practice · Hard

CH 4 · TOPIC 9 4 marks

13 A pattern is made from two **similar** trapeziums.



Show that the shaded area is 294 cm^2

(4 marks)

Q22**PRACTICE**

Practice · Very Hard

CH 4 · TOPIC 9 4 marks

- 4 The total surface area of a solid hemisphere is equal to the curved surface area of a cylinder.

The radius of the hemisphere is r cm.

The radius of the cylinder is twice the radius of the hemisphere.

Given that

$$\text{volume of hemisphere} : \text{volume of cylinder} = 1 : m$$

find the value of m .

(4 marks)

Q23**PRACTICE**

Practice · Very Hard

CH 4 · TOPIC 9 4 marks

- 6 A standard tin and a large tin are mathematically similar.
The **volume** of the large tin is 50% more than the volume of the standard tin.
Both tins are cylinders.
The radius of the standard tin is 10cm.
Calculate the radius of the large tin.

..... cm

(4 marks)

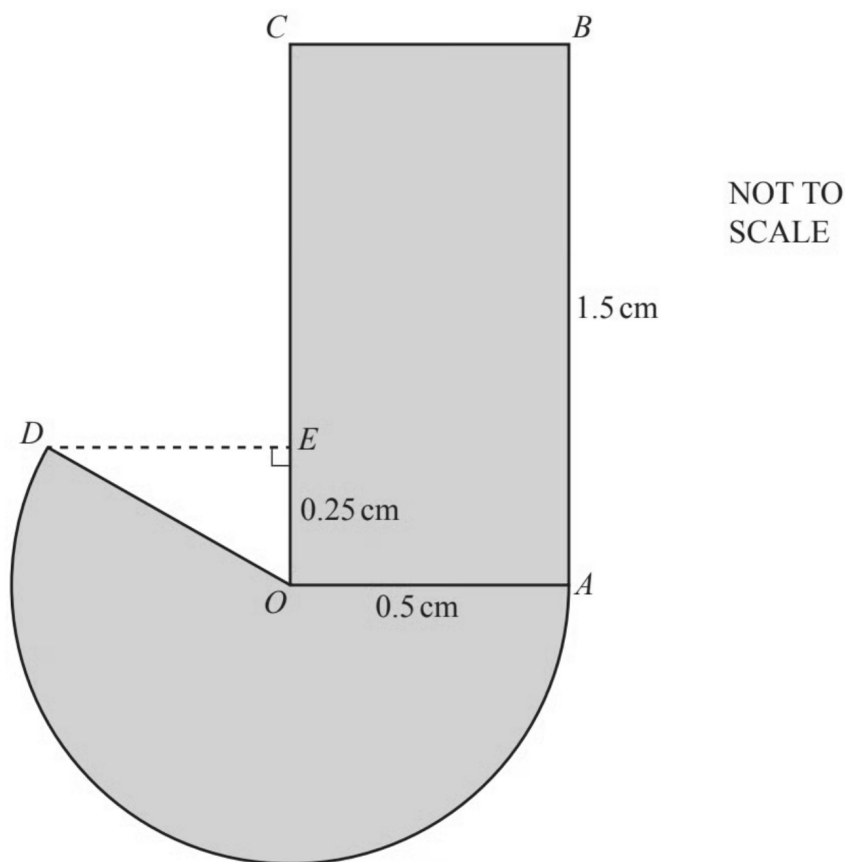
Q24**PRACTICE**

Practice · Very Hard

CH 4 · TOPIC 9

11 marks

8 (a)



The diagram shows a company logo made from a rectangle and a major sector of a circle. The circle has centre O and radius OA . $OA = OD = 0.5$ cm and $AB = 1.5$ cm. E is a point on OC such that $OE = 0.25$ cm and angle $OED = 90^\circ$.

Calculate the perimeter of the logo.

..... cm

(5 marks)

(b) Calculate the area of the logo.

..... cm^2

(3 marks)

(c) A mathematically similar logo is drawn. The area of this logo is 77.44 cm^2 .

Calculate the radius of the major sector in this logo.

..... cm

(3 marks)

PAST PAPERS

Q25 PAST PAPER Edexcel 4MA1 · January 2020 · Paper 1HR · Q20

CH 4 · TOPIC 9 5 marks

20 The diagram shows a frustum of a cone and a sphere.

The frustum is made by removing a small cone from a large cone.
The cones are similar.

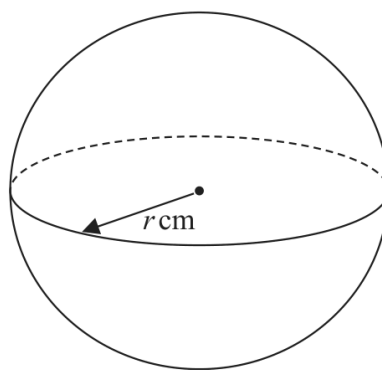
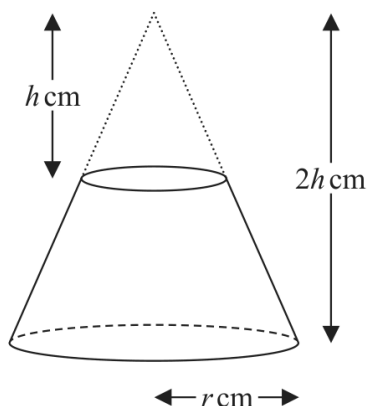


Diagram **NOT**
accurately drawn

The height of the small cone is h cm.
The height of the large cone is $2h$ cm.
The radius of the base of the large cone is r cm.

The radius of the sphere is r cm.

Given that the volume of the frustum is equal to the volume of the sphere,

find an expression for r in terms of h .

Give your expression in its simplest form.

$r = \dots\dots\dots$

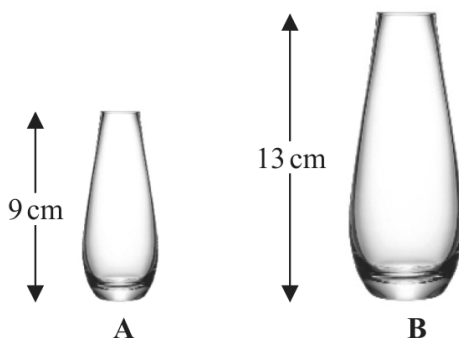
(Total for Question 20 is 5 marks)

Q26**PAST PAPER**

Edexcel 4MA1 · January 2020 · Paper 2HR · Q20

CH 4 · TOPIC 9 4 marks

20 The diagram shows two similar vases, **A** and **B**.



The height of vase **A** is 9 cm and the height of vase **B** is 13 cm.

Given that

$$\text{surface area of vase A} + \text{surface area of vase B} = 1800 \text{ cm}^2$$

calculate the surface area of vase **A**.

..... cm²

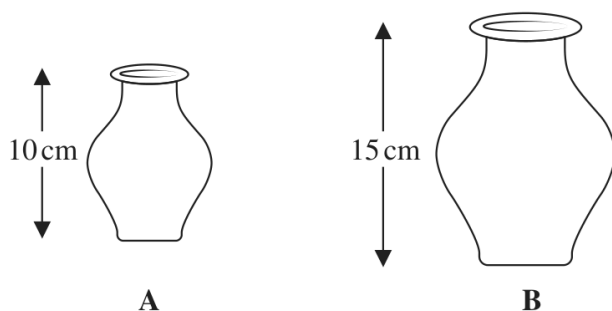
(Total for Question 20 is 4 marks)

Q27**PAST PAPER**

Edexcel 4MA1 · January 2022 · Paper 1H · Q17

CH 4 · TOPIC 9 4 marks

17 **A** and **B** are two similar vases.



Vase **A** has height 10 cm.

Vase **B** has height 15 cm.

The difference between the volume of vase **A** and the volume of vase **B** is 1197 cm^3

Calculate the volume of vase **A**

..... cm^3

(Total for Question 17 is 4 marks)

Q28**PAST PAPER**

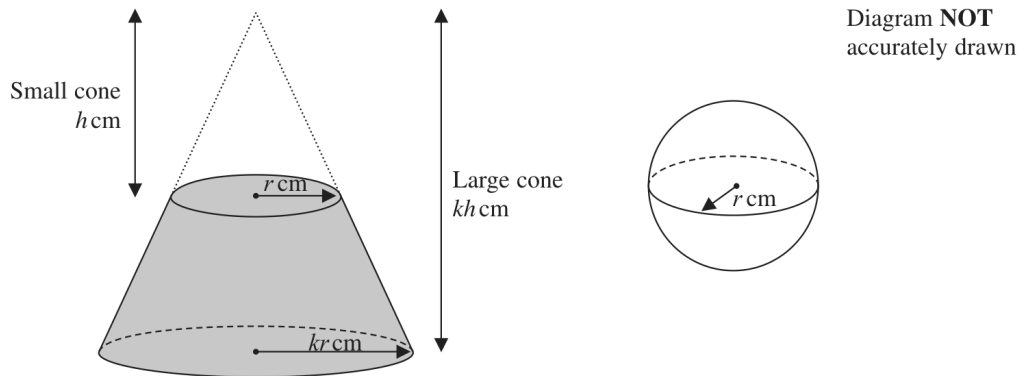
Edexcel 4MA1 · June 2022 · Paper 2HR · Q24

CH 4 · TOPIC 9 6 marks

24 The diagram shows a frustum of a cone, and a sphere.

The frustum, shown shaded in the diagram, is made by removing the small cone from the large cone.

The small cone and the large cone are similar.



The height of the small cone is h cm and the radius of the base of the small cone is r cm.
The height of the large cone is kh cm and the radius of the base of the large cone is kr cm.
The radius of the sphere is r cm.

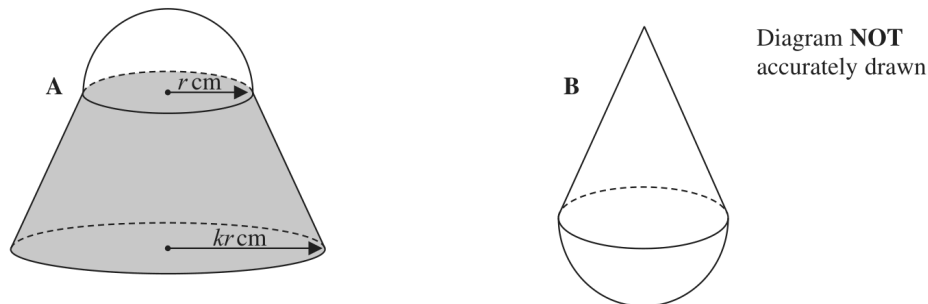
The sphere is divided into two hemispheres, each of radius r cm.

Solid **A** is formed by joining one of the hemispheres to the frustum.

The plane face of the hemisphere coincides with the upper plane face of the frustum, as shown in the diagram below.

Solid **B** is formed by joining the other hemisphere to the small cone that was removed from the large cone.

The plane face of the hemisphere coincides with the plane face of the base of the small cone, as shown in the diagram below.



The volume of solid **A** is 6 times the volume of solid **B**.

Given that $k > \sqrt[3]{7}$

find an expression for h in terms of k and r

$h = \dots\dots\dots$

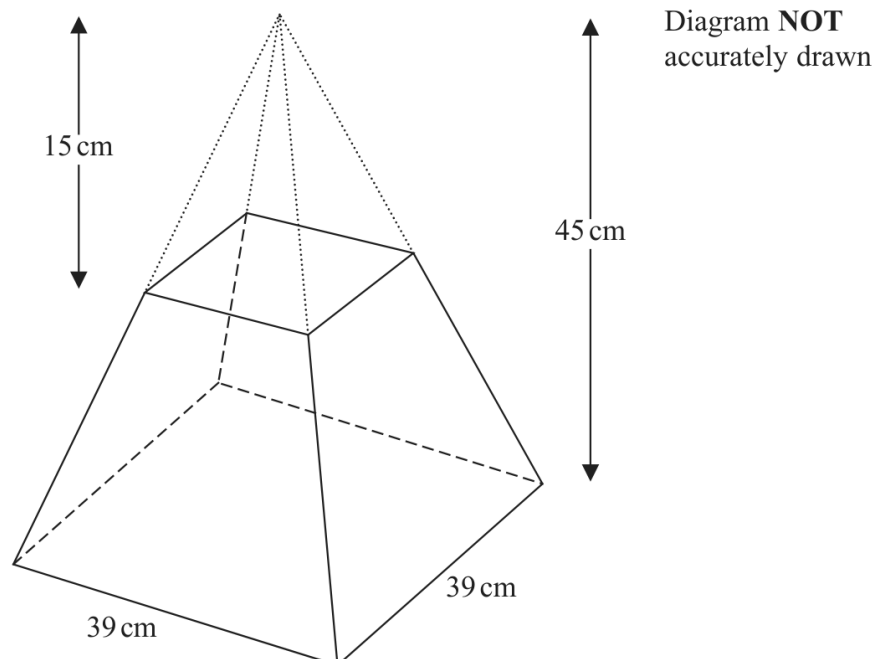
(Total for Question 24 is 6 marks)

Q29**PAST PAPER**

Edexcel 4MA1 · May 2023 · Paper 1H · Q23

CH 4 · TOPIC 9 5 marks

- 23 A frustum is made by removing a small square-based pyramid from a similar large squared-based pyramid as shown in the diagram.



The height of the small pyramid is 15 cm.

The height of the large pyramid is 45 cm.

The square base of the large pyramid has side length 39 cm.

Work out the **total** surface area of the frustum.

Give your answer correct to the nearest whole number.

..... cm²

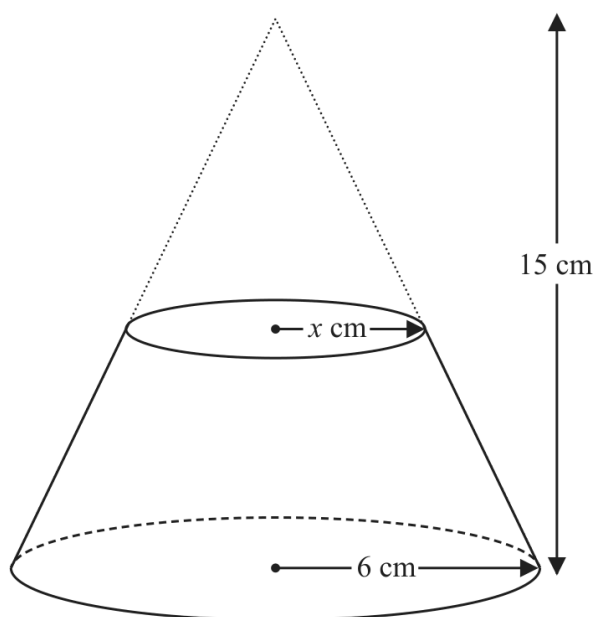
(Total for Question 23 is 5 marks)

Q30**PAST PAPER**

Edexcel 4MA1 · November 2023 · Paper 2H · Q23

CH 4 · TOPIC 9 5 marks**23** Here is a frustum of a cone.

The frustum is made by removing a small cone from a similar large cone.

Diagram **NOT**
accurately drawn

The height of the large cone is 15 cm.

The radius of the base of the large cone is 6 cm.

The radius of the base of the small cone is x cm.Given that the volume of the frustum is $\frac{4212}{25}\pi \text{ cm}^3$ work out the value of x

Show clear algebraic working.

 $x = \dots\dots\dots$ **(Total for Question 23 is 5 marks)**

Q31**PAST PAPER**

Edexcel 4MA1 • November 2023 • Paper 1H • Q23

CH 4 • TOPIC 9 5 marks**23** Solid **A** is similar to solid **B**Here is some information about solid **A** and solid **B**

	solid A	solid B
Height (cm)	3^x	
Area (cm ²)	7776	486
Volume (cm ³)	8^x	2^{x+4}

Work out the height of solid **B**

Give your answer as a decimal.

..... cm

(Total for Question 23 is 5 marks)

Q32**PAST PAPER**

Edexcel 4MA1 · May 2024 · Paper 1HR · Q18

CH 4 · TOPIC 9 4 marks

18 The diagram shows two similar vases, **A** and **B**

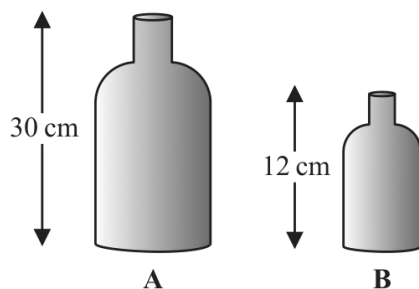


Diagram **NOT**
accurately drawn

The height of vase **A** is 30 cm

The height of vase **B** is 12 cm

Given that

$$\text{surface area of vase A} - \text{surface area of vase B} = 178.5 \text{ cm}^2$$

find the surface area of vase **A**

..... cm²

(Total for Question 18 is 4 marks)

Q33**PAST PAPER**

Edexcel 4MA1 · June 2025 · Paper 2H · Q26

CH 4 · TOPIC 9 4 marks

26 **A** and **B** are two similar solids.

The height of solid **A** is 31 cm

The height of solid **B** is 18.6 cm

Given that

$$\text{volume of solid A} - \text{volume of solid B} = 735 \text{ cm}^3$$

find the volume of solid **A**

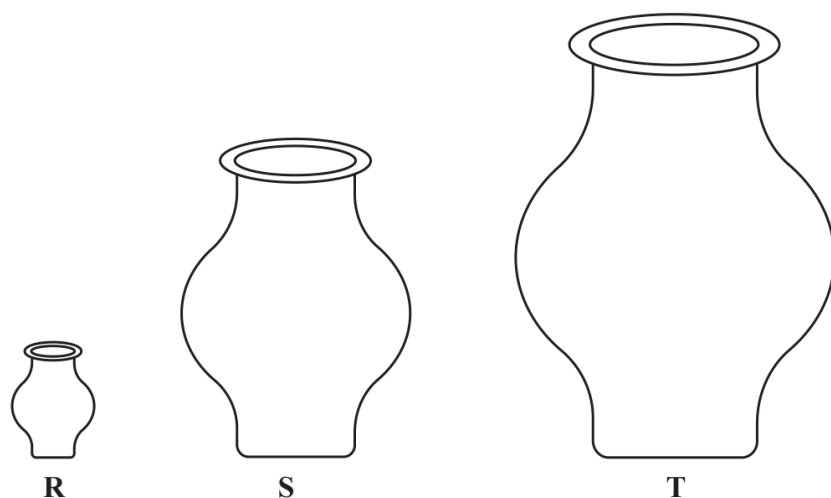
..... cm³

(Total for Question 26 is 4 marks)

Q34 PAST PAPER Edexcel 4MA1 · June 2025 · Paper 1HR · Q26

CH 4 · TOPIC 9 4 marks

26 R, S and T are three similar vases.

Diagram **NOT**
accurately drawnThe volume of vase **S** is 72.8% more than the volume of vase **R**The height of vase **R** is h cmThe height of vase **T** is $6h$ cmThe surface area of vase **S** is A cm²The surface area of vase **T** is kA cm²Work out the value of k $k = \dots\dots\dots$ **(Total for Question 26 is 4 marks)**

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|-------------------------|---|
| 01 | Scale factors | Length k , area k^2 , volume k^3 . |
| 02 | One from another | Root back to k , then the power you need. |
| 03 | Harder | Sum / difference: V and k^3V ; frustum = big – small. |

BOTTOM LINE Back to the length factor first → square for area → cube for volume.

Answers

Area and Volume of Similar Shapes • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • LENGTH, AREA AND VOLUME SCALE FACTORS

- Q1** Dr Eslam • Area from length $12 \times \left(\frac{5}{2}\right)^2 = 75 \text{ cm}^2$
- Q2** Practice • Medium $k = 6: 300 \times 36 = 10\,800 \text{ cm}^2$
- Q3** Practice • Medium curved surfaces 9 : 25; heights 3 : 5
- Q4** Practice • Medium $\frac{90 \times 10^6}{72^3} = 241 \text{ cm}^3$
- Q5** Practice • Hard (a) $AB = 7.5 \text{ cm}$ (b) $36 \times 1.5^2 - 36 = 45 \text{ cm}^2$
- Q6** Practice • Hard areas 1 : 16 : 25: shaded 15, unshaded 1 + 9 = 10: 3 : 2
- Q7** Practice • Very Hard $\sqrt{73.5 \div 6} = 3.5: 7 : 2$

IDEA 02 • FIND ONE FROM ANOTHER

Q8	Dr Eslam • Volume from area	$k = \frac{3}{2}: 40 \times \frac{27}{8} = 135 \text{ cm}^3$
Q9	Practice • Medium	$928 \times \left(\frac{405}{960}\right)^{2/3} = 522 \text{ cm}^2$
Q10	Practice • Medium	$k = \frac{7}{4}: 176 \times \frac{49}{16} = 539 \text{ cm}^2$
Q11	Practice • Hard	$k = \frac{2}{3}: 405 \times \frac{8}{27} = 120$
Q12	Practice • Hard	$k = \frac{2}{3}: 297 \times \frac{4}{9} = 132$
Q13	Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q20	625 cm^3
Q14	Edexcel 4MA1 • January 2021 • Paper 1H • Q20	630 cm^2
Q15	Edexcel 4MA1 • May 2021 • Paper 2H • Q20	518.4 cm^3
Q16	Edexcel 4MA1 • January 2022 • Paper 1HR • Q18	$10 : 21$
Q17	Edexcel 4MA1 • June 2022 • Paper 1HR • Q20	$k = 13.8$
Q18	Edexcel 4MA1 • June 2024 • Paper 2HR • Q24	(a) $k = 2.5$ (b) $d = 64$
Q19	Edexcel 4MA1 • November 2025 • Paper 2H • Q23	$672 \times 1.5^3 = 2268 \text{ cm}^3$

IDEA 03 • HARDER PROBLEMS: DIFFERENCES, FRUSTUMS, ALGEBRA

- Q20** Dr Eslam • Difference $V(1.5^3 - 1) = 570$: $V = 240 \text{ cm}^3$
- Q21** Practice • Hard large 350; small $k = 0.4$: $350 \times 0.16 = 56$:
 $350 - 56 = 294$
- Q22** Practice • Very Hard $3\pi r^2 = 4\pi r h \Rightarrow h = \frac{3}{4}r$: $m = 4.5$
- Q23** Practice • Very Hard $10 \times \sqrt[3]{1.5} = 11.4 \text{ cm}$
- Q24** Practice • Very Hard (a) 5.83 cm (b) 1.21 cm^2 (c) $k = 8$: radius 4 cm
- Q25** Edexcel 4MA1 • January 2020 • Paper 1HR • Q20 $\frac{7}{8} \cdot \frac{2}{3}\pi r^2 h = \frac{4}{3}\pi r^3$: $r = \frac{7h}{16}$
- Q26** Edexcel 4MA1 • January 2020 • Paper 2HR • Q20 583.2 cm^2
- Q27** Edexcel 4MA1 • January 2022 • Paper 1H • Q17 504 cm^3
- Q28** Edexcel 4MA1 • June 2022 • Paper 2HR • Q24 $h = \frac{10r}{k^3 - 7}$
- Q29** Edexcel 4MA1 • May 2023 • Paper 1H • Q23 5090 cm^2
- Q30** Edexcel 4MA1 • November 2023 • Paper 2H • Q23 $x = \frac{12}{5}$
- Q31** Edexcel 4MA1 • November 2023 • Paper 1H • Q23 $k = 4, x = 5$: $243 \div 4 = 60.75 \text{ cm}$
- Q32** Edexcel 4MA1 • May 2024 • Paper 1HR • Q18 212.5 cm^2
- Q33** Edexcel 4MA1 • June 2025 • Paper 2H • Q26 937.5 cm^3
- Q34** Edexcel 4MA1 • June 2025 • Paper 1HR • Q26 $k = 25$